Chapter 8: Convex Functions

Convex Analysis and Monotone Operator Theory in Hilbert Spaces

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Chapter 8: Convex Functions

Definition 8.1: A function $f : \mathcal{H} \to [-\infty, +\infty]$ is *convex* if

$$f(\alpha x + (1-\alpha)y) \leq \alpha f(x) + (1-\alpha)f(y)$$

for all $x, y \in \text{dom} f$ and $\alpha \in [0, 1]$.

Definition 8.7: f is *strictly convex* if

$$x \neq y \implies f(\alpha x + (1-\alpha)y) < \alpha f(x) + (1-\alpha)f(y)$$

for all $x, y \in \text{dom} f$ and $\alpha \in (0, 1)$.

Key Concept: The epigraph $\text{epi} f = \{(x, r) \in \mathcal{H} \times \mathbb{R} : f(x) \leq r\}$ is convex *if and only if* f is convex.

Proposition 8.11: Let $f : \mathcal{H} \to [-\infty, +\infty]$ be convex. Then

$$f\left(\sum_{i\in I} \alpha_i x_i\right) \leq \sum_{i\in I} \alpha_i f(x_i)$$

for all finite families $(\alpha_i)_{i\in I}$ in $[0,1]$ with $\sum_{i\in I} \alpha_i = 1$ and $(x_i)_{i\in I}$ in $\operatorname{dom} f$.

Corollary 8.12: For convex f: if $(x_i)_{i\in I}$ are in $\operatorname{dom} f$ with weights $(\alpha_i)_{i\in I}$ summing to 1, then Jensen's inequality holds.

Interpretation: The function value at a weighted average is at most the weighted average of function values.

Examples: Basic Convex Functions

Example 8.9: The norm $\|\cdot\| : \mathcal{H} \to \mathbb{R}$ is convex:

$$\|\alpha x + (1-\alpha)y\| \leq \alpha\|x\| + (1-\alpha)\|y\|$$

but *not strictly convex* (e.g., take $x = -y$).

Example 8.10: The squared norm $\|\cdot\|^2 : \mathcal{H} \to \mathbb{R}$ is *strictly convex*. Proof: By Corollary 2.15 (parallelogram law).

Example 8.19: For a bounded sequence $(z_n)_{n\in\mathbb{N}}$ in $\mathcal{H}$:

$$f : \mathcal{H} \to \mathbb{R}, \quad f(x) = \limsup_{n\to\infty} \|x - z_n\|$$

is convex and Lipschitz continuous with constant 1.

Convex Functions on Subsets

Definition: Let C be a nonempty subset of $\operatorname{dom} f$. Then f is *convex on* C if

$$(\forall x, y \in C)(\forall \alpha \in [0,1]) \quad f(\alpha x + (1-\alpha)y) \leq \alpha f(x) + (1-\alpha) f(y)$$

Remark 8.8: $f + \iota_C$ is convex if and only if C is convex and f is convex on C.

Corollary 8.5: For convex $f : \mathcal{H} \to [-\infty, +\infty]$ and every $\xi \in \mathbb{R}$:

$$\operatorname{lev}_{\xi} f := \{x \in \mathcal{H} : f(x) \leq \xi\}$$

is a convex set.

Convexity and Differentiability

Proposition 8.14: Let $\phi : \mathbb{R} \to [-\infty, +\infty]$ be differentiable on a nonempty open interval I in domϕ. Then:

1. If ϕ' is increasing on I, then ϕ is convex on I.
2. If ϕ' is strictly increasing on I, then ϕ is strictly convex on I.

Intuition: Increasing derivative $\Rightarrow$ increasing slope $\Rightarrow$ convex function.

Convex Function Examples from Calculus

Example 8.15: Let $\psi : \mathbb{R} \to \mathbb{R}$ be increasing and $\alpha \in \mathbb{R}$. Define

$$\phi : \mathbb{R} \to [-\infty, +\infty], \quad \phi(x) = \begin{cases} \int_\alpha^x \psi(t)dt & \text{if } x \geq \alpha \\ +\infty & \text{otherwise} \end{cases}$$

Then ϕ is convex. If ψ is strictly increasing, then ϕ is strictly convex.

Example 8.22: For continuous increasing $\psi : \mathbb{R} \to \mathbb{R}$ with $\psi(0) \geq 0$:

$$f : \mathcal{H} \to \mathbb{R}, \quad f(x) = \int_0^{\|x\|} \psi(t)dt$$

is convex. If ψ is strictly increasing, f is strictly convex.

Power Functions and p-norms

Example 8.23: For $p \in [1, +\infty)$, the function $\| \cdot \|^p$ is strictly convex:

$$(\forall x, y \in \mathcal{H}) \quad \|x + y\|^p \leq 2^{p-1}(\|x\|^p + \|y\|^p)$$

Example 8.31: The Rényi divergence between $x = (\xi_i)_{i \in I} \in \mathbb{R}^N_+$ and $y = (\eta_i)_{i \in I} \in \mathbb{R}^N_+$ is:

$$f(x, y) = \begin{cases} \sum_{i \in I} \xi_i^\alpha \eta_i^{-\alpha} & \text{if } x \in \mathbb{R}^N_{++}, y \in \mathbb{R}^N_{++} \\ +\infty & \text{otherwise} \end{cases}$$

For $\alpha \in [1, +\infty)$, this is convex (apply Example 8.26).

Preservation under Supremum and Addition

Proposition 8.16: Let $(f_i)_{i \in I}$ be a family of convex functions from $\mathcal{H}$ to $[-\infty, +\infty]$. Then

$$\sup_{i \in I} f_i \text{ is convex}$$

Proposition 8.17: The set of convex functions from $\mathcal{H}$ to $[-\infty, +\infty]$ is closed under addition and multiplication by strictly positive real numbers.

Proposition 8.18: Let $(f_n)_{n \in \mathbb{N}}$ be a sequence of convex functions from $\mathcal{H}$ to $[-\infty, +\infty]$ such that $(f_n)_{n \in \mathbb{N}}$ is pointwise convergent. Then $\lim f_n$ is convex.

Composition with Increasing Convex Functions

Proposition 8.21: Let $f : \mathcal{H} \to [-\infty, +\infty]$ and $\phi : \mathbb{R} \to [-\infty, +\infty]$ be convex functions. Set $C = \mathrm{conv}(\mathbb{R} \cap \mathrm{ran}\, f)$ and extend ϕ to $\tilde{\phi} : [-\infty, +\infty] \to [-\infty, +\infty]$ by setting $\tilde{\phi}(+\infty) = +\infty$. Suppose that $C \subseteq \mathrm{dom}\phi$ and that ϕ is increasing on C. Then

$$\tilde{\phi} \circ f \text{ is convex}$$

Key Example: If f is convex and $\phi(t) = e^t$, then $e^{f(x)}$ is convex.

Proposition 8.24: Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and $\mathcal{H} = L^2((\Omega, \mathcal{F}, \mu); \mathbb{R})$. Let $\varphi : \mathcal{H} \to [-\infty, +\infty]$ be measurable and convex such that

$$(\forall x \in \mathcal{H})(\exists \rho \in L^1((\Omega, \mathcal{F}, \mu); \mathbb{R})) \quad \varphi \circ x \geq \rho \quad \mu\text{-a.e.}$$

Then the integral function

$$f : \mathcal{H} \to [-\infty, +\infty], \quad f(x) = \int_\Omega \varphi(x(\omega))\mu(d\omega)$$

is convex.

Perspective Functions

Definition & Proposition 8.25: Let $\varphi : \mathcal{H} \to [-\infty, +\infty]$ be convex, set $C = \{1\} \times \operatorname{epi}\varphi$, and define

$$f : \mathbb{R} \times \mathcal{H} \to [-\infty, +\infty], \quad f(\xi, x) = \begin{cases} \xi\varphi(x/\xi) & \text{if } \xi > 0 \\ +\infty & \text{otherwise} \end{cases}$$

Then $\operatorname{epi} f = \operatorname{cone} C$ and f is convex.

Application: This construction shows how to homogenize a convex function.

Important Divergences (Examples 8.26–8.31)

Csiszár ϕ-divergence: For convex $\varphi : \mathbb{R} \to [-\infty, +\infty]$,

$$d_\varphi(x, y) = \sum_{i \in I} \eta_i \varphi(\xi_i/\eta_i) \quad \text{is convex in } (x, y)$$

Kullback-Leibler: $d_{KL}(x, y) = \sum_i \xi_i \ln(\xi_i/\eta_i) - \xi_i + \eta_i$

Jeffreys: $d_J(x, y) = \sum_i (\xi_i - \eta_i) \ln(\xi_i/\eta_i)$

Pearson Chi-squared: $d_P(x, y) = \sum_i (\xi_i - \eta_i)^2/\eta_i$

Hellinger: $d_H(x, y) = \sum_i (\sqrt{\xi_i} - \sqrt{\eta_i})^2$

Rényi ($\alpha \in [1, +\infty)$): $d_R(x, y) = \sum_i \xi_i^\alpha \eta_i^{-\alpha}$

Marginal and Minkowski Gauge Functions

Proposition 8.35: Let $\mathcal{K}$ be a real Hilbert space and $F : \mathcal{H} \times \mathcal{K} \to [-\infty, +\infty]$ be convex. Then the marginal function

$$f : \mathcal{H} \to [-\infty, +\infty], \quad f(x) = \inf_{\mathcal{K}} F(x, \mathcal{K})$$

is convex.

Definition & Example 8.36: The *Minkowski gauge* of a convex set C with $0 \in C$ is

$$m_C : \mathcal{H} \to [-\infty, +\infty], \quad m_C(x) = \inf\{\xi \in \mathbb{R}_{++} : x \in \xi C\}$$

This is convex. Moreover: $m_C(0) = 0$ and $m_C(\lambda x) = \lambda m_C(x)$ for $\lambda \in \mathbb{R}_{++}$.

Lipschitz Continuity of Convex Functions

Proposition 8.37: Let $f : \mathcal{H} \to [-\infty, +\infty]$ be proper and convex, let $x_0 \in \operatorname{dom} f$, and let $\rho \in \mathbb{R}_{++}$. Then:

1. If $\eta = \sup f(B(x_0; \rho)) < +\infty$ and $\alpha \in [0, 1]$, then

$$(\forall x \in B(x_0; \alpha\rho)) \quad |f(x) - f(x_0)| \leq \alpha(\eta - f(x_0))$$

2. If $\delta = \operatorname{diam} f(B(x_0; 2\rho)) < +\infty$, then f is Lipschitz continuous relative to $B(x_0; \rho)$ with constant δ/ρ.

Interpretation: Convex functions are automatically Lipschitz on bounded sets where they are bounded.

Main Continuity Result

Theorem 8.38: Let $f : \mathcal{H} \to [-\infty, +\infty]$ be proper and convex, and let $x_0 \in \operatorname{dom} f$. Then the following are equivalent:

1. f is locally Lipschitz continuous near x_0.
2. f is continuous at x_0.
3. f is bounded on a neighborhood of x_0.
4. f is bounded above on a neighborhood of x_0.

Moreover, if one of these conditions holds, then f is locally Lipschitz continuous on int dom f.

Intuition: For convex functions, very weak boundedness assumptions imply strong continuity properties.

Continuity in Finite Dimensions

Corollary 8.40: Suppose $\mathcal{H}$ is finite-dimensional and $f : \mathcal{H} \to \mathbb{R}$ is convex. Then f is continuous.

Corollary 8.41: Suppose $\mathcal{H}$ is finite-dimensional, $f : \mathcal{H} \to [-\infty, +\infty]$ is proper and convex, and C is a nonempty closed bounded subset of ri dom f. Then f is Lipschitz continuous relative to C.

Key Principle: In finite-dimensional spaces, proper convex functions are always continuous on the relative interior of their domain.

Interior of Sublevel Sets

Proposition 8.46: Let $f : \mathcal{H} \to [-\infty, +\infty]$. Then:

1. If f is upper semicontinuous at $x_0 \in \mathcal{H}$ with $f(x_0) < 0$, then $x_0 \in \operatorname{int} \operatorname{lev}_{\leq 0} f$.
2. If f is convex and there exists $x_0 \in \mathcal{H}$ with $f(x_0) < 0$, then $\operatorname{int} \operatorname{lev}_{\leq 0} f = \operatorname{lev}_{<0} f$.

Corollary 8.47: Let $f : \mathcal{H} \to [-\infty, +\infty]$ be convex such that $\operatorname{lev}_{<0} f \neq \emptyset$. Suppose one of:

1. f is upper semicontinuous on $\operatorname{lev}_{<0} f$.
2. f is lower semicontinuous and $\operatorname{dom} f$ is open.
3. $\mathcal{H}$ is finite-dimensional and $\operatorname{dom} f$ is open.

Then $\operatorname{int} \operatorname{lev}_{<0} f = \operatorname{lev}_{<0} f$.

The Huber Loss Function

Example 8.44: For $\rho \in \mathbb{R}_{++}$, the Huber function is

$$f : \mathbb{R} \to \mathbb{R}, \quad f(x) = \begin{cases} \rho|x| - \frac{\rho^2}{2} & \text{if } |x| > \rho \\ \frac{|x|^2}{2} & \text{if } |x| \leq \rho \end{cases}$$

Then f is continuous and convex.

Application: The Huber function is a robust loss function in machine learning that combines quadratic loss (near zero) with linear loss (far from zero).

Core of a Convex Function

Definition: The *core* of a convex function f is

$$\text{core } f = \{x \in \mathcal{H} : f(x) < +\infty \text{ and } f \text{ is finite on a neighborhood of } x\}$$

Corollary 8.39: Let $f : \mathcal{H} \to [-\infty, +\infty]$ be proper and convex. Suppose one of:

1. f is bounded above on some neighborhood.
2. f is lower semicontinuous.
3. $\mathcal{H}$ is finite-dimensional.

Then $\text{cont } f = \text{int dom } f$.

Interpretation: Under mild conditions, the "interior of continuity" coincides with the "interior of the domain."

Example 1: Convexity Verification in Python

```python
import numpy as np

# Verify Jensen's inequality: f(alpha*x + (1-alpha)*y) <= alpha*f(x) + (1-alpha)*f(y)
def verify_convexity(f, x, y, alpha):
    """Verify Jensen's inequality for function f at points x, y"""
    left_side = f(alpha * x + (1 - alpha) * y)
    right_side = alpha * f(x) + (1 - alpha) * f(y)
    return left_side <= right_side + 1e-10

# Test with f(x) = x^2 (strictly convex)
f = lambda x: x**2
x, y = np.array([1.0, 2.0]), np.array([3.0, 4.0])

for alpha in [0.3, 0.5, 0.7]:
    is_convex = verify_convexity(f, x, y, alpha)
    print(f"alpha={alpha}: Jensen satisfied = {is_convex}")
```

Output: All alphas satisfy the inequality with equality when f is strictly convex.

Example 2: Kullback-Leibler Divergence

```python
import numpy as np

def kl_divergence(p, q):
    """Kullback-Leibler divergence KL(p || q) = sum(p_i ln(p_i/q_i))"""
    return np.sum(p * np.log(p / q))

def kl_convexity_check(p1, q1, p2, q2, alpha):
    """Check convexity of KL divergence as function of first argument"""
    # Convex combination of distributions
    p_mixed = alpha * p1 + (1 - alpha) * p2
    q_avg = alpha * q1 + (1 - alpha) * q2

    left = kl_divergence(p_mixed, q_avg)
    right = alpha * kl_divergence(p1, q1) + (1-alpha) * kl_divergence(p2, q2)

    return left <= right + 1e-10

# Test with probability distributions
p1 = np.array([0.2, 0.3, 0.5])
p2 = np.array([0.1, 0.4, 0.5])
q = np.array([0.25, 0.25, 0.5])

alpha = 0.6
is_convex = kl_convexity_check(p1, q, p2, q, alpha)
print(f"KL divergence convexity verified: {is_convex}")
```

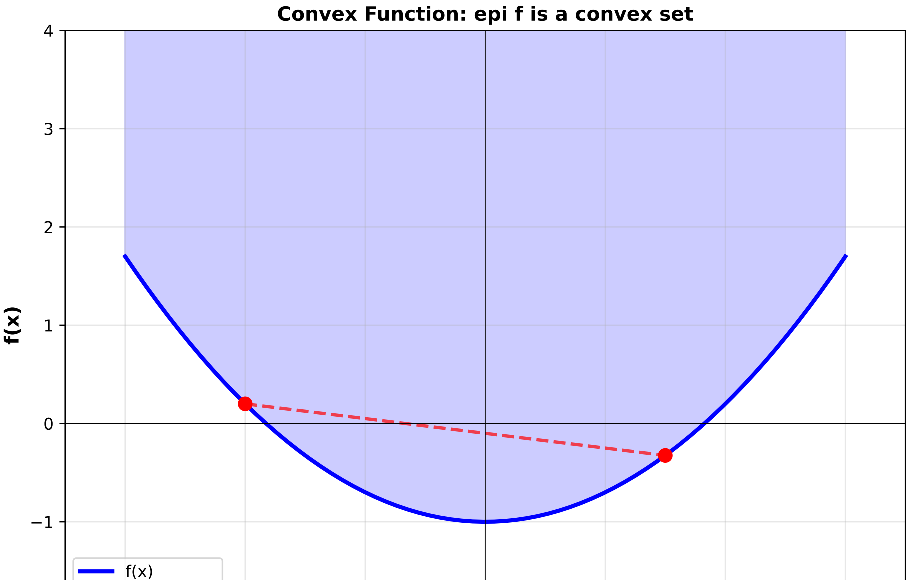
Example 3: Lipschitz Continuity

```python
import numpy as np

def verify_lipschitz(f, x1, x2, L):
    """Verify Lipschitz continuity: |f(x1) - f(x2)| <= L * ||x1 - x2||"""
    func_diff = abs(f(x1) - f(x2))
    norm_diff = np.linalg.norm(x1 - x2)

    return func_diff <= L * norm_diff + 1e-10

# Test with f(x) = ||x|| (Lipschitz with constant 1)
f_norm = lambda x: np.linalg.norm(x)
x1 = np.array([1.0, 2.0, 3.0])
x2 = np.array([1.5, 2.5, 2.5])

L = 1.0  # Lipschitz constant for norm
is_lipschitz = verify_lipschitz(f_norm, x1, x2, L)
print(f"Norm is Lipschitz continuous with L={L}: {is_lipschitz}")

# For bounded region, check ||x||^2
f_squared = lambda x: np.linalg.norm(x)**2
L_bounded = 2 * max(np.linalg.norm(x1), np.linalg.norm(x2))
is_lipschitz_sq = verify_lipschitz(f_squared, x1, x2, L_bounded)
print(f"Squared norm is Lipschitz on bounded region: {is_lipschitz_sq}")
```

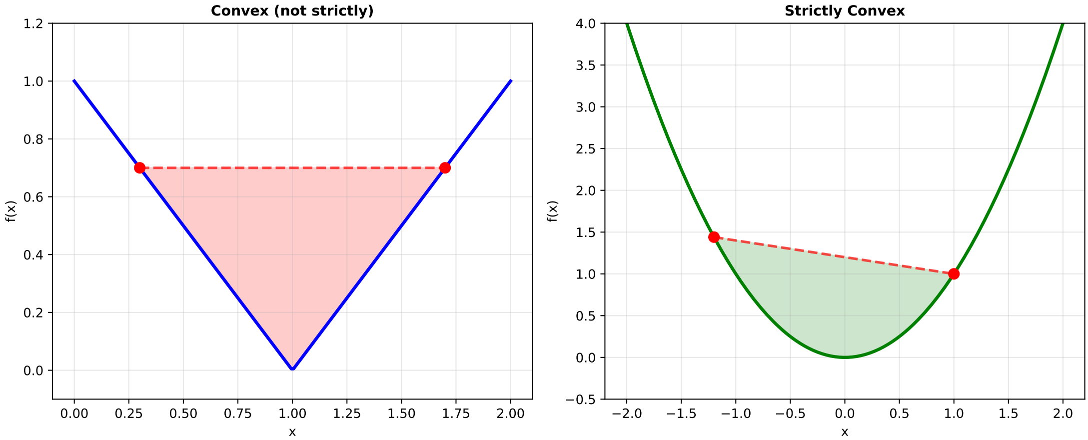
Example 4: Minkowski Gauge

```python
import numpy as np

def minkowski_gauge(point, convex_set):
    """Compute Minkowski gauge: m_C(x) = inf{lambda > 0 : x in lambda*C}"""
    # For a ball of radius r centered at origin:
    # m_C(x) = ||x|| / r

    # For unit ball:
    r_set = 1.0  # radius of convex set
    return np.linalg.norm(point) / r_set

def verify_gauge_homogeneity(x, lambda_val):
    """Verify: m_C(lambda*x) = lambda * m_C(x)"""
    gauge_x = minkowski_gauge(x, None)
    gauge_lambda_x = minkowski_gauge(lambda_val * x, None)

    return abs(gauge_lambda_x - lambda_val * gauge_x) < 1e-10

# Test
x = np.array([1.0, 2.0])
lambda_val = 2.5

is_homogeneous = verify_gauge_homogeneity(x, lambda_val)
print(f"Gauge homogeneity verified: {is_homogeneous}")
print(f"m_C(x) = {minkowski_gauge(x, None):.4f}")
print(f"m_C({lambda_val}*x) = {minkowski_gauge(lambda_val*x, None):.4f}")
```

Convex Function Definition

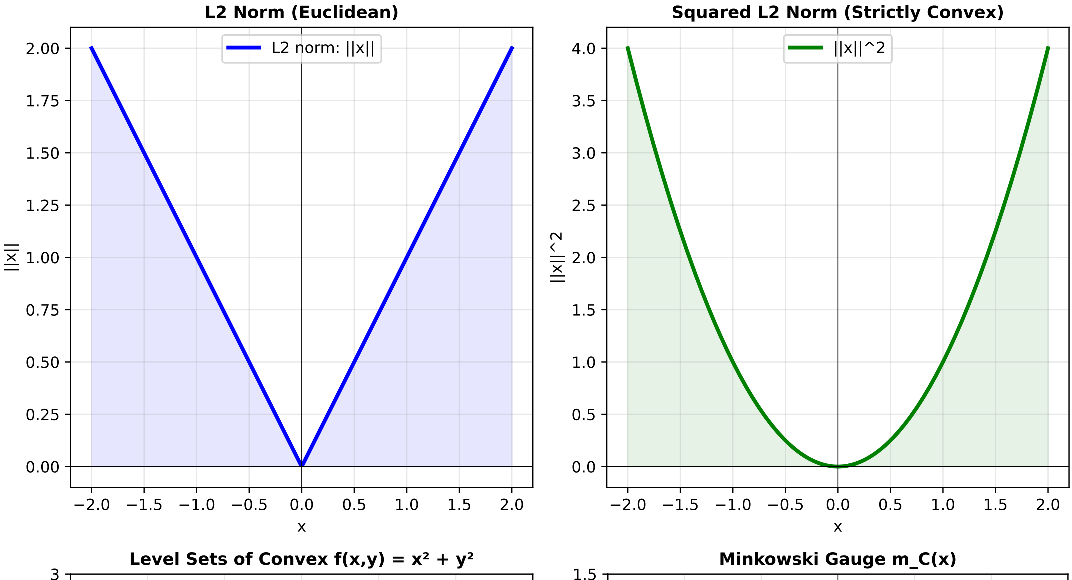


Strictly Convex vs. Convex

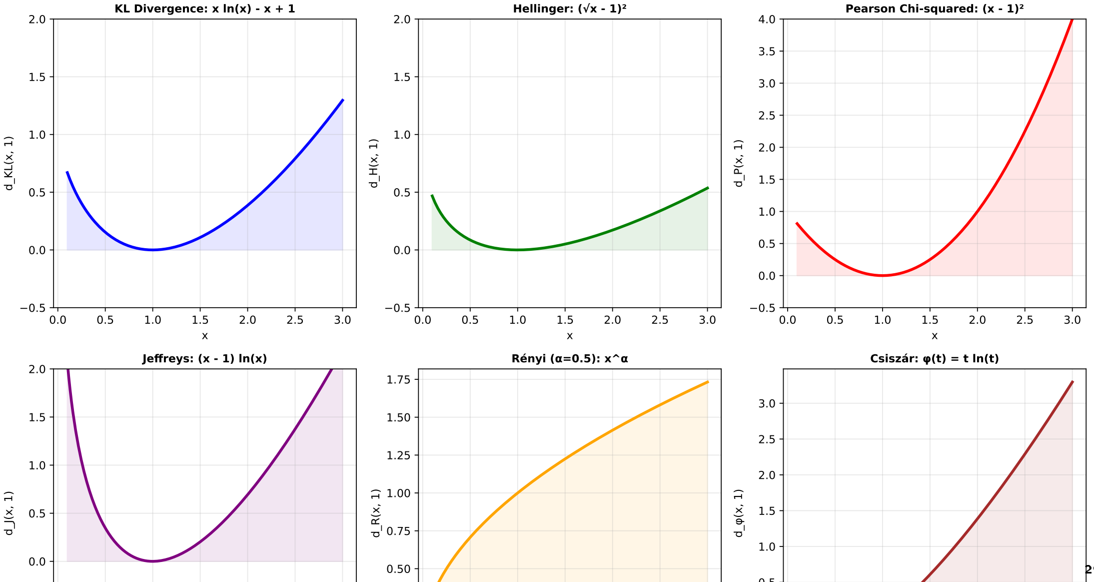


Left: Convex but not strictly (linear segment in graph). Right: Strictly convex (strict inequality).

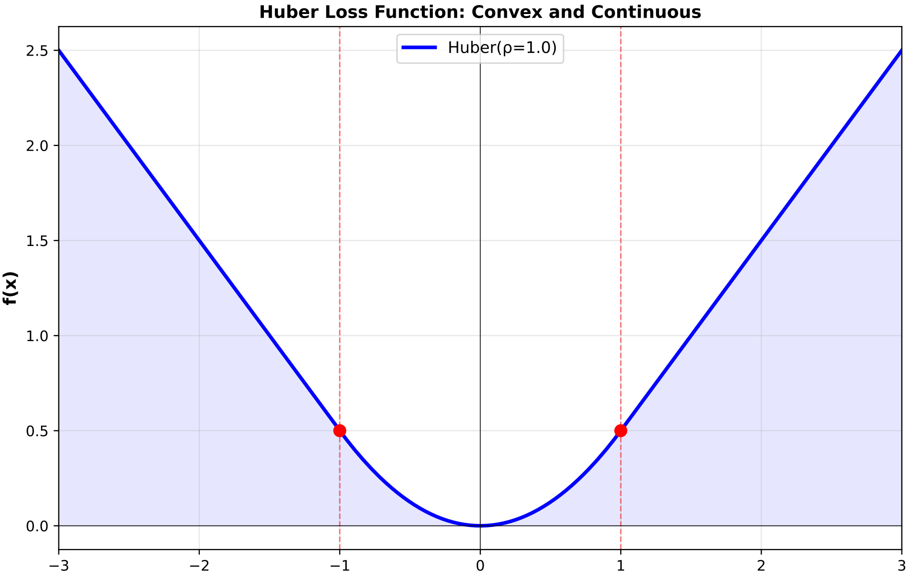
Norms and Level Sets



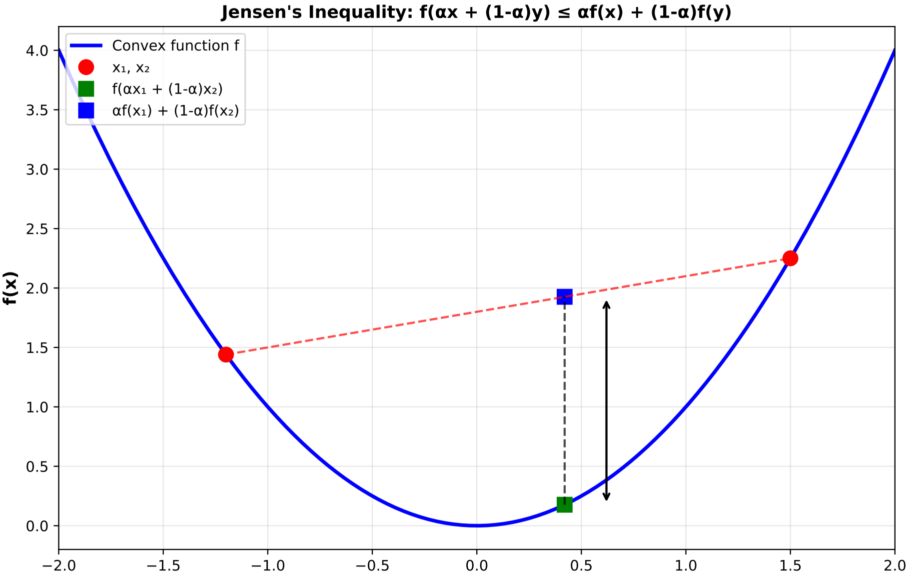
Important Divergence Functions



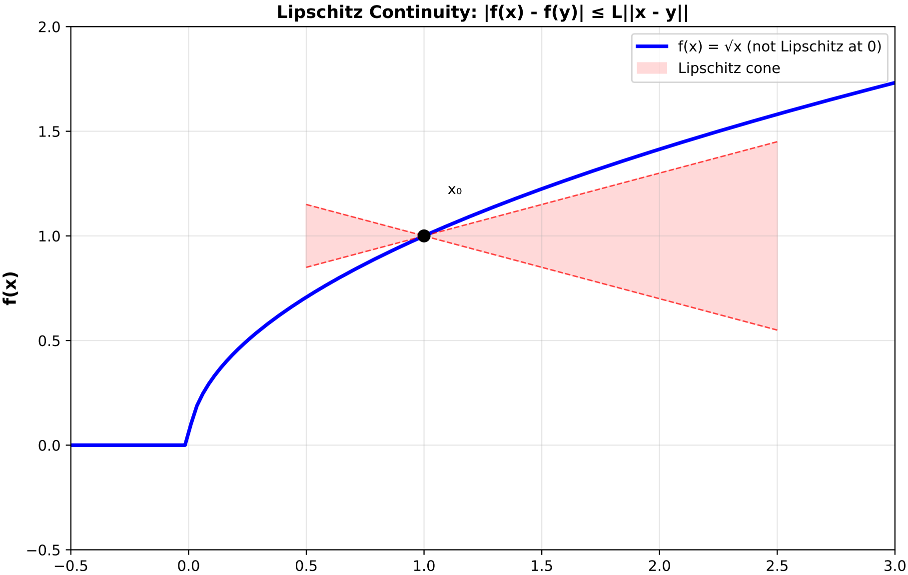
Huber Loss Function



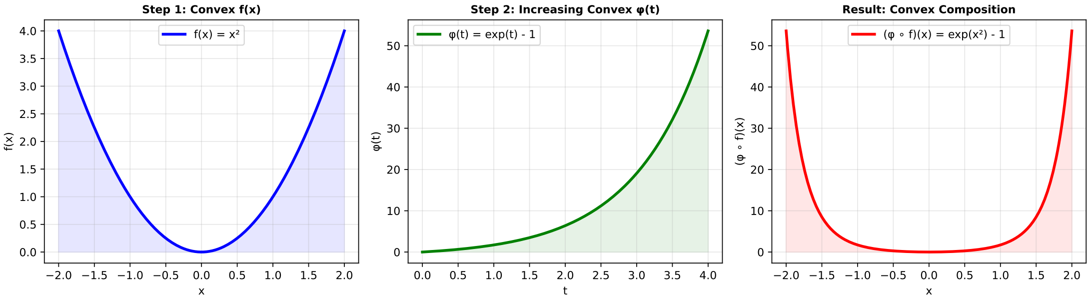
Jensen's Inequality Illustration



Lipschitz Continuity



Composition of Convex Functions



Composition of inner convex function f with increasing convex function ϕ gives convex result.

Core Definitions:

- Convex functions and their epigraphs
- Strictly convex functions
- Properties of level sets (convex sets)

Preservation of Convexity:

- Supremum of convex functions is convex
- Pointwise limits of convex functions are convex
- Composition with increasing convex functions preserves convexity
- Integrals and perspective functions preserve convexity

Topological Properties:

- Boundedness implies Lipschitz continuity
- Theorem 8.38: Equivalence of four continuity conditions
- In finite dimensions: proper convex functions are continuous on int dom f

Key Inequalities and Examples

Jensen's Inequality:

$$f\left(\sum_i \alpha_i x_i\right) \le \sum_i \alpha_i f(x_i)$$

Norms are convex: $\|\alpha x + (1-\alpha)y\| \le \alpha\|x\| + (1-\alpha)\|y\|$

Squared norms are strictly convex

Important Divergences:

- Kullback-Leibler: $\sum_i p_i \ln(p_i/q_i)$
- Jeffreys: $\sum_i (p_i - q_i)\ln(p_i/q_i)$
- Hellinger: $\sum_i (\sqrt{p_i} - \sqrt{q_i})^2$

All are convex in their arguments!

Applications and Connections

Optimization:

- Convex functions have unique global minima (if convex constraints)
- Lower semicontinuous convex functions have minimizers in finite-dim spaces
- Subdifferential calculus applies (see later chapters)

Statistics and Machine Learning:

- Kullback-Leibler divergence: information theory
- Huber loss: robust regression
- Convex functions: convex optimization problems

Functional Analysis:

- Properties of proper convex functions lead to subdifferentials
- Connection to weak* lower semicontinuity

Reading Guide

Essential Results to Master:

1. Definition 8.1: Convex functions (via convexity of epigraph)
2. Proposition 8.11: Jensen's inequality
3. Proposition 8.16–8.18: Preservation under operations
4. Theorem 8.38: Main continuity theorem
5. Corollary 8.39–8.41: Finite-dimensional consequences

Recommended Exercises:

- Exercise 8.1–8.5: Basic convexity properties
- Exercise 8.10: Arithmetic-geometric mean inequality
- Exercise 8.14: Sublevel sets and convexity
- Exercise 8.18–8.22: Advanced applications

Next Chapter (Chapter 9): Lower Semicontinuous Convex Functions